

AMS Bootcamp 2026: Bash, Git, Python

Brendan McKinley

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 - Writing and executing Bash scripts?
 - Python scripting and Jupyter notebooks?
- How many have set up a GitHub account?

Agenda

- 1 Bash: CLI
- 2 Version Control with Git
- 3 Bash: Scripting
- 4 Python/Jupyter Notebooks

Interacting with the OS

- Shell provides a text- or UI-based interface with the underlying operating system of your machine
 - Defaults
 - Linux/macOS: BASH (Bourne-Again Shell)
 - Windows: PowerShell/Command Prompt
 - BASH can be used on Windows with Git Bash -
 - <https://git-scm.com/install>

Bash commands and keywords

- `..` – Refers to parent directory
- `ls` – List contents of current directory
- `cd` – Change directory
- `mkdir` – Create directory
- `mv` – Move a file (or rename a file, if moving to the same directory)
- `scp` – Used to securely copy files (and directories, when using the `-r` flag) between machines

- Install guide: <https://helpcenter.mines.edu/TDClient/1946/Portal/KB/ArticleDet?ID=154280>

Integrated Development Environment (IDE)

- Centralized platform for development, debugging, testing, and version control
- Examples: Eclipse, RStudio, Matlab, Visual Studio, Visual Studio Code
- VSCode: Lightweight, versatile, good shapes and colors

Exercise: Install VSCode

- ① Navigate to <https://code.visualstudio.com/> and install Visual Studio Code.
- ② Familiarize yourself with the UI.
 - Top drop-down menu
 - File → Open, File → Open Folder
 - Terminal → New Terminal
 - View → Command Palette
 - Left-hand menu
 - LHS: Explorer, Search, Source Control

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- ➍ Collaboration and code reviews for shared work

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- 1 Backing up your own work
- 2 Logically organizing your own work
- 3 Keeping track of your work's history
- 4 Collaboration and code reviews for shared work
- 5 Automated testing for shared work

Basic Git workflow

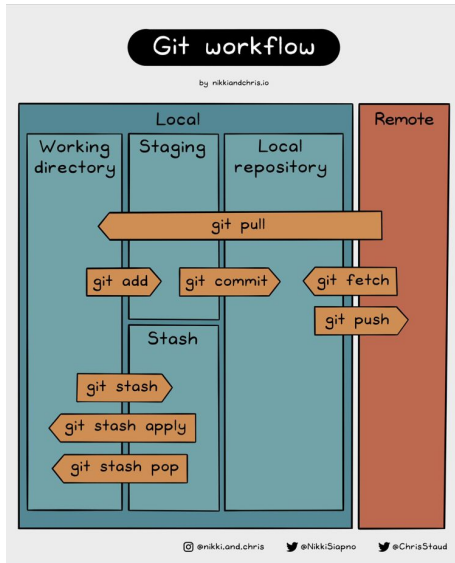


Figure: Typical Git workflow overview.

Principles of good version control

- Descriptive commit names. Like, actually
- “Commit early, commit often” but also commit consistently
- Organize by importance – tagging, feature branches, releases

Git commands

- Cheat sheet: <https://git-scm.com/cheat-sheet>
- `git init`: Initialize new repository
- `git add <file>`: Stage changes for commit
- `git commit -m "message"`: Commit staged changes with message "message"
- `git stash`: Store changes in the stash
- `git stash pop`: Restore most recently stashed changes
- `git branch`: List, create, delete branches

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- I use the `diff` features on GitHub's web UI and VSCode constantly
- GitHub's web UI allows you to see the history of commits affecting a specific file or folder

Exercise: Install Git

- 1 Navigate to `https://git-scm.com/install` and install Git for your OS.
- 2 Once Git is installed, run `git --version` in your terminal to verify the installation.

Exercise: Clone a repository

- 1 If you don't already have one, create a new GitHub account.
- 2 Navigate to
`https://github.com/brendan-e-mckinley/bootCampAMS/`
- 3 Clone via the terminal with `git clone`
`https://github.com/brendan-e-mckinley/bootCampAMS.git`
 - On the GitHub web UI, the “Code” dropdown gives you several options for cloning a repository.
 - You can open a terminal directly from VSCode.
- 4 Open the repository in VSCode.

Exercise: Make a new branch

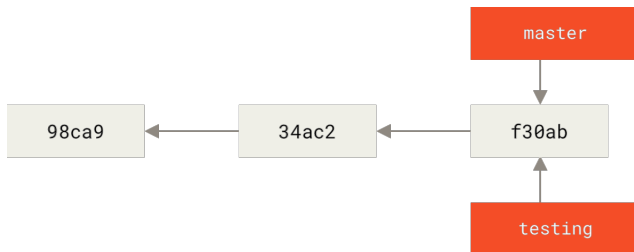


Figure: The result of running `git branch testing`. Credit: <https://git-scm.com/book/en/v2/Git-Branching-Branches-in-a-Nutshell>

- 1 In your terminal, run `git checkout -b <firstname>_<lastname>`
- 2 Verify that the message "Switched to a new branch '`<firstname>_<lastname>`'" is returned
- 3 Push your new branch using the VSCode UI
 - Or via the terminal: `git push --set-upstream origin <firstname>_<lastname>`

Exercise: .gitignore

- 1 Update the .gitignore file so that the repository excludes any files with the .png extension.
- 2 Add a .png file to the repository.
- 3 Ensure the image is excluded from the “Source Control” tab in VSCode.
- 4 Commit and push your update to the .gitignore file to your branch.

Exercise: Open a pull request

- 1 Navigate to `https://github.com/brendan-e-mckinley/bootCampAMS/`
- 2 Go to the “Branches” tab
- 3 Select the drop-down menu for your branch and click “New pull request”

Exercise: Create a Repository

- 1 Go to your account page on GitHub.
- 2 Select the “Repositories” tab.
- 3 Select “New”.
- 4 Name the repository and give it a description.
- 5 Notice the visibility, README, and `.gitignore` settings.

Break

Python scripting

- Structure

- Whitespace matters!
- Main entry point

```
if __name__ == "__main__":
```

- Function definition

```
def greater_than(arg_1, arg_2):
```

- Variable assignments

```
arg_1 = 10  
arg_2 = 1
```

Exercise: Running a Python script

- ❶ If you haven't already, install Python.
 - ❶ Download the relevant Python installer from <https://www.python.org/downloads/>
 - ❷ Add Python to PATH if you're installing on a Windows machine.
- ❷ Run a script from the terminal.
 - ❶ Navigate to the src directory of the bootCampAMS repository.
 - ❷ Execute the `greater.py` Python script using the default Python installation with `python greater.py`
- ❸ Run a script from VSCode.
 - ❶ Open the `greater.py` Python script in VSCode.
 - ❷ Open the Command Palette with "View" → "Command Palette" or `Cmd + Shift + P`
 - ❸ Click "Python: Select Interpreter" and select the default interpreter for VSCode.
 - ❹ Click "Run Python File" in the upper-right corner of the VSCode UI.

Exercise: Debugging a Python script

- 1 Open the `greater.py` Python script in VSCode.
- 2 Place a breakpoint on the `greater_than()` function call.
- 3 Select "Python Debugger: Debug Python File" from the "▶" dropdown in the upper-right corner.
- 4 Step into the `greater_than()` function.
- 5 Hover over the `arg_2` variable to see its contents.
 - Notice the "Run and Debug" information on the left-hand side.

Bash scripting

- You can automate any command line Bash commands by organizing them in a Bash script.
- Useful for:
 - Repeated operations
 - Logically connecting and sharing information between different Bash commands
 - Communicating with third party applications like a GitHub CI/CD pipeline

Exercise: Running a bash script

- 1 Open the `execute_operation.sh` script in the `src` directory of the `bootCampAMS` repository in Visual Studio Code.
 - Notice the shebang followed by the Bash path: `#!/bin/bash`
- 2 Navigate to the `src` directory in the terminal.
- 3 Make the script executable with `chmod +x execute_operation.sh`
 - More information on `chmod`:
<https://www.redhat.com/en/blog/introduction-chmod>
- 4 Execute the script with `./execute_operation.sh`

Aside: What's wrong with `operation.py`?

- Learn more: <https://peps.python.org/pep-0008/>

Virtual environments

- Package: a folder with some Python code, along with everything you need to use that code in your projects
- Certain packages will be useful for many different Python projects
 - matplotlib
 - numpy
- Others will only be relevant for specific projects
- Virtual environments allow you to customize package installations for specific projects
- Learn more: <https://docs.python.org/3/library/venv.html>

Exercise: Setting up a virtual environment

- 1 In your terminal, create a new virtual environment with `python -m venv boot_camp`.
- 2 Activate the environment:
 - BASH (Linux/macOS): `source boot_camp/bin/activate`
 - Command Prompt (Windows): `boot_camp\Scripts\activate.bat`
- 3 Ensure that the terminal now lists `(boot_camp)` before the current active directory.
- 4 Install `pytest` in your virtual environment with `pip install pytest`.

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 - SOLID design principles –
<https://realpython.com/solid-principles-python/>

Exercise: Python testing

- 1 Open a terminal in VSCode and navigate to the top-level directory of the repository (`/bootcampAMS`).
- 2 Ensure that your `(boot_camp)` virtual environment is activated.
- 3 Run `python -m pytest`.

Exercise: Conda environment from file

- 1 Conda is a third-party package and environment manager.
- 2 Install Conda – <https://docs.conda.io/projects/conda/en/latest/user-guide/install/index.html>
 - Or, if you hate that, you're valid
 - Should have everything you need to create a virtual environment from which you can run the notebook
 - Required dependencies are listed in `environment.yml`
- 3 Once Conda is installed, you can build a new virtual environment from the `environment.yml` file with `conda env create -f environment.yml`

Jupyter notebooks

- Combine Markdown, Python code, and plots in a single readable notebook.
- Useful for:
 - Caching and memory
 - Mathematical notation and Latex
 - Helping others get a deeper understanding of shared code
 - Presenting findings
 - Organizing code for your own benefit

Exercise: Run a Jupyter notebook

- 1 In the top-right corner of `notebook.ipynb` in VSCode, choose the `jupyter-env` Python kernel.
- 2 Run each section of the Jupyter notebook in order. Notice:
 - Run/debug cell
 - Edit cell – markdown cells interpret Latex code directly
 - Split cell